

# Highlights

Sieve: Training-Free, Model-Free, Question-Conditioned Context  
Compression for Efficient Long-Context LLM Inference: An  
Accuracy–Compression–Cost Frontier Study of Model-Free Information  
Selection

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- A model-free, training-free selector compresses long contexts in milliseconds.
- Three-dataset, 1600-record study maps the accuracy–compression–cost frontier.
- BM25 leads raw recall; the positional prior trades recall for reader-side placement.
- A measured GPT-2 perplexity gate costs 255–320× more compute for less recall.
- Evidence recall correlates only weakly with end-task answer accuracy.